

Assignment (2)

1.	a)	How is a UDP socket fully identified? What about a TCP socket? What is the difference between the full identification of both sockets?	04
	b)	Describe why an application developer might choose to run an application over UDP rather than TCP.	03
	c)	Why is it that voice and video traffic is often sent over TCP rather than UDP in today's Internet?	03
2.	Suppose Host A sends two TCP segments back to back to Host B over a TCP connection. The first segment has sequence number 90; the second has sequence number 110.		10
	a)	How much data is in the first segment?	
	b)	Suppose that the first segment is lost but the second segment arrives at B. In the acknowledgment that Host B sends to Host A, what will be the acknowledgment number?	
	c)	Is it possible for an application to enjoy reliable data transfer even when the application runs over UDP? If so, how?	
	d)	Suppose a process in Host C has a UDP socket with port number 6789. Suppose both Host A and Host B each send a UDP segment to Host C with destination port number 6789. Will both of these segments be directed to the same socket at Host C? If so, how will the process at Host C know that these two segments originated from two different hosts?	
3.	a)	In our rdt protocols, why did we need to introduce sequence numbers?	10
	b)	why did we need to introduce timers?	
	c)	Suppose that the roundtrip delay between sender and receiver is constant and known to the sender. Would a timer still be necessary in protocol rdt 3.0, assuming that packets can be lost? Explain.	
	d)	In protocol rdt3.0, the ACK packets flowing from the receiver to the sender do not have sequence numbers (although they do have an ACK field that contains the sequence number of the packet they are acknowledging). Why is it that our ACK packets do not require sequence numbers?	
4.	a)	Visit the Go-Back-N interactive animation at the Companion Website. a. Have the source send five packets, and then pause the animation before any of the five packets reach the destination. Then kill the first packet and resume the animation. Describe what happens. b. Repeat the experiment, but now let the first packet reach the destination and kill the first acknowledgment. Describe again what happens. c. Finally, try sending six packets. What happens?	10
	b)	Repeat R12, but now with the Selective Repeat interactive animation. How are Selective Repeat and Go-Back-N different?	
5.	a)	Consider two hosts, A and B, transmitting a large file to a server C, over a bottleneck link with rate R. To transfer the file, the hosts use TCP with	10

	the same parameters (including MSS and RTT) and start their transmissions at the same time. Host A uses a single TCP connection for the entire file, while Host B uses 9 simultaneous TCP connections, each for a portion (i.e., a chunk) of the file. What is the overall transmission rate achieved by each host at the beginning of the file transfer? Is this situation fair? b) Suppose Client A requests a web page from Server S through HTTP and its socket is associated with port 33000. a. What are the source and destination ports for the segments sent from A to S? b. What are the source and destination ports for the segments sent from S to A? c. Can Client A contact to Server S using UDP as the transport protocol? d. Can Client A request multiple resources in a single TCP connection?	
6.	a) UDP and TCP use 1s complement for their checksums. Suppose you have the following three 16 bit words: 0101001101100110; 0111010010110100; 0000110111000001. What is the 1s complement of the sum of these words? Show all work. Why is it that UDP offers a checksum? With the 1's complement scheme, how does the receiver detect errors? Describe how a single bit flip can be detected. b) Assume that a host receives a UDP segment with 01011101 11110010 (we separated the values of each byte with a space for clarity) as the checksum. The host adds the 16-bit words over all necessary fields excluding the checksum and obtains the value 00110010 00001101. Is the segment considered correctly received or not? What does the receiver do?	10
7.	a) Suppose that the UDP receiver computes the Internet checksum for the received UDP segment and finds that it matches the value carried in the checksum field. Can the receiver be absolutely certain that no bit errors have occurred? Explain. b) In protocol rdt3.0, the ACK packets flowing from the receiver to the sender do not have sequence numbers (although they do have an ACK field that contains the sequence number of the packet they are acknowledging). Why is it that our ACK packets do not require sequence numbers? c) Draw the FSM for the receiver side of protocol rdt3.0.	10
8.	a) Consider a channel that can lose packets but has a maximum delay that is known. Modify protocol rdt2.1 to include sender timeout and retransmit. Informally argue why your protocol can communicate correctly over this channel. b) Consider the rdt2.2 receiver in Figure 3.14, and the creation of a new packet in the self-transition (i.e., the transition from the state back to itself) in the Wait-for-0-from-below and the Wait-for-1-from-below states:	10

	<code>sndpkt=make_pkt(ACK,1,checksum)</code> and <code>sndpkt=make_pkt(ACK,0,checksum)</code>. Would the protocol work correctly if this action were removed from the self-transition in the Wait-for-1-from-below state? Justify your answer. What if this event were removed from the self-transition in the Wait-for-0-from-below state?	
9.	a) Assume Host A is streaming a video from Server B using UDP. Also assume that the network suddenly becomes very congested while Host A is seeing the video. Is there any way to handle this situation with UDP? What about with TCP? Is there any other option? b) Consider a stop-and-wait data-transfer protocol that provides error checking and retransmissions but uses only negative acknowledgments. Assume that negative acknowledgments are never corrupted. Would such a protocol work over a channel with bit errors? What about over a lossy channel with bit errors?	10
10.	a) Consider the GBN protocol with a sender window size of 4 and a sequence number range of 1,024. Suppose that at time t, the next in-order packet that the receiver is expecting has a sequence number of k. Assume that the medium does not reorder messages. Answer the following questions: a. What are the possible sets of sequence numbers inside the sender's window at time t? Justify your answer. b. What are all possible values of the ACK field in all possible messages currently propagating back to the sender at time t? Justify your answer. b) Give one example where buffering out-of-order segments would significantly improve the throughput of a GBN protocol. c) Consider a scenario where the three hosts A, B, and C are connected as a ring: A to B, B to C, and C to A. Assume that A and C run protocol rdt3.0, whereas B simply relays all messages received from A to C. a. Does this arrangement enable reliable delivery of messages from A to C? b. Can B tell if a certain message has been correctly received by A?	10